M-ADSAI-ZM002-26

Requirements Verification

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Functional Requirements Evaluation

Table 1 – Perception and local processing functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-01	Person Detection	The system shall detect a person within the Reachy Mini's camera view within 5 seconds after the person enters the camera view during an exercise session.	Yes
Explanation:		The system detects a person within the Reachy Mini's camera view within 5 seconds after the person enters the camera view during an exercise session.	
FR-02	Local Processing	The system shall process captured camera data entirely on the Reachy Mini or the connected local computer, without requiring external processing for core activity-tracking functionality.	Yes
Explanation:		The captured camera data is processed on the Reachy Mini Robot and does therefore not leave the robot.	
FR-03	Person Tracking	The system shall track the detected person continuously while the person remains visible within the camera's field of view during an exercise session.	Yes
Explanation:		The robot tracks the user while the exercises are performed.	
FR-05	Pose Estimation	The system shall provide the input representation required by the selected activity recognition approach, which may be an explicit pose/skeletal representation or image/video data.	Yes
Explanation:		The system provides the input representation required by the selected activity recognition, using skeletal representation.	
FR-18	User Detection Loss	The system shall detect that the user is no longer visible to the camera within 2 seconds after the user leaves the camera's field of view during an exercise session.	Yes
Explanation:		The system detects that the user is no longer visible to the camera within 2 seconds after the user leaves the camera's field of view during an exercise session.	

Table 2 – Robot behaviour and interaction functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-04	User Following	The system shall move the Reachy Mini's head to follow the detected user while the user remains within the camera's field of view during an exercise session.	Yes
Explanation:		The Reachy Mini robot follows the user for the exercises by moving its head.	
FR-12	Audio Feedback	The system shall provide audio feedback through the Reachy Mini's speaker whenever an audio feedback response is triggered.	Yes
Explanation:		The robot provides audio feedback when an exercise is performed.	
FR-13	Head Movement Feedback	The system shall use movements of the Reachy Mini's head to provide feedback to the user whenever a head-movement feedback response is triggered.	Yes
Explanation:		If an exercise is performed successfully the Reachy Mini will do a short dance with its head and antennas.	
FR-14	Antenna Movement Feedback	The system shall use movements of the Reachy Mini's antennas to provide feedback to the user whenever an antenna-movement feedback response is triggered.	Yes
Explanation:		If an exercise is performed successfully the Reachy Mini will do a short dance with its head and antennas.	

Table 3 – Activity and exercise recognition functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-06	Static Movement Recognition	The system shall recognize all static positions listed as supported activities during an exercise session.	Yes
Explanation:		The system is able to detect how long the user performs the arms up exercise.	
FR-07	Dynamic Movement Recognition	The system shall recognize all dynamic movements listed as supported activities during an exercise session.	Yes
Explanation:		The system can recognize chair rise and pick-up exercises.	
FR-08	Physiotherapy Exercise Recognition	The system shall recognize all physiotherapy exercises listed as supported exercises during an exercise session.	Yes
Explanation:		The system can recognize chair rise, pick-up and arms up exercises.	

Table 4 – Assesment and feedback functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-09	Movement Assessment	The system shall assess the execution of supported physiotherapy exercises using the predefined movement criteria for each exercise.	Yes
Explanation:		The system is able to detect length of an exercise (arms up) and if an exercise is performed (Chair rise and pick-up).	
FR-10	Repetition Detection	The system shall detect completed repetitions according to the predefined repetition criteria during an exercise session.	Yes
Explanation:		The system can recognize chair rise, pick-up and arms up exercises in repetitions.	

Table 5 – Session management functional requirements

ID	Title	Requirement	Requirement met by artifact
FR-11	Session Management	The system shall allow the user to start and end an exercise session.	Yes
Explanation:		The system allows the user to start and end an exercise session.	
FR-12	Exercise Monitoring	The system shall monitor the user's movements continuously throughout an active exercise session.	Yes
Explanation:		The system monitors the user throughout the exercise session.	
FR-13	Exercise Result	The system shall provide the user with the detected exercise results and assessment results at the end of an exercise session.	Yes
Explanation:		The Reachy Mini tells the user, the results after an exercise session and when an exercise is detected.	

Non-Functional Requirements Evaluation

Table 6 – Non-Functional Requirements

<table>
<tr><th>ID</th><th>Title</th><th>Requirement</th><th>Requirement met by artifact</th></tr>
<tr><td>NFR-01</td><td>Privacy</td><td>The system shall ensure that captured video and session data remain on the Reachy Mini and/or the connected local computer for the entire duration of system operation and shall not be transmitted to external systems.</td><td>Yes</td></tr>
<tr><td colspan="2">Explanation:</td><td colspan="2">The video and session data stays on the Reachy Mini</td></tr>
<tr><td>NFR-02</td><td>Recognition Accuracy</td><td>The system shall provide activity recognition for the supported activities when evaluated using the predefined test dataset. The recognition performance shall be measured and reported.</td><td>Yes</td></tr>
<tr><td colspan="2">Explanation:</td><td colspan="2">The system provides activity recognition for the supported activities when evaluated using the predefined test dataset. The recognition performance is measured and shown in figure 1.
<table><tr><th>Class</th><th>Full body</th><th>Waist-up</th></tr><tr><td>none</td><td>0.902</td><td>0.855</td></tr><tr><td>wave</td><td>0.918</td><td>0.873</td></tr><tr><td>squat</td><td>0.922</td><td>0.803</td></tr><tr><td>clapping</td><td>0.898</td><td>0.899</td></tr><tr><td>jumping_jacks</td><td>0.887</td><td>0.861</td></tr><tr><td>pushup</td><td>0.381</td><td>0.137</td></tr></table>Figure 1: per-fold F1 score</td></tr>
<tr><td>NFR-03</td><td>Reliability</td><td>The system shall complete an exercise session without requiring an application restart under normal operating conditions.</td><td>Yes</td></tr>
<tr><td colspan="2">Explanation:</td><td colspan="2">The system completes an exercise session without requiring an application restart under normal operating conditions.</td></tr>
<tr><td>NFR-04</td><td>Usability</td><td>The system shall allow first-time users to start an exercise session without assistance from the development team.</td><td>Yes</td></tr>
<tr><td colspan="2">Explanation:</td><td colspan="2">The system allows first-time users to start the session without assistance.</td></tr>
<tr><td>NFR-05</td><td>Safety</td><td>The system shall not perform unexpected or uncontrolled robot movements during normal operation.</td><td>Yes</td></tr>
<tr><td colspan="2">Explanation:</td><td colspan="2">The Robot does not perform unexpected or uncontrolled robot movements during normal operation.</td></tr>
</table>

Conclusion

Requirements Verification has shown that all functional and non-functional requirements identified in the SRS have been satisfied in the Reachy Mini prototype. The system has the capability to detect and track users, recognize physiotherapy exercises that are supported, monitor exercise execution and give feedback via audio and robot gestures. Also, the prototype has satisfied the requirements for processing locally, privacy protection, reliability, usability and safety. The recognition accuracy has been tested with the use of a predefined test dataset.

However, despite the fact that all the requirements have been met, there is still need for additional testing and development to make sure that the system will work properly in a healthcare setting. The recognition accuracy should be tested on a greater number of patients and different environments. Generally, the prototype is a good basis for further development into a system which can conduct CMAS assessments at home.